# Monte Carlo Methods

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Generating Samples
Markov Chain Monte Carlo + Discrepancy
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Selected Topics

**Git website and repository**
**Canvas**

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# Improving Efficiency

Owen, Chapters 8–10, 15–17

Test 2 Oct 29

**MATH 565 Monte Carlo Methods, Fred Hickernell, Fall 2025**

# Variance Reduction

$$\mu = \mathbb{E}[f(\boldsymbol{X})] = \mathbb{E}[g(\boldsymbol{X})], \quad \boldsymbol{X} \sim \mathcal{U}[0,1]^d$$

Is it better to use

$$\widehat{\mu}_{f,n} = \frac{1}{n}\sum_{i=0}^{n-1} f(\boldsymbol{X}_i) \quad \text{or} \quad \widehat{\mu}_{g,n} = \frac{1}{n}\sum_{i=0}^{n-1} g(\boldsymbol{X}_i), \qquad \boldsymbol{X}_0, \boldsymbol{X}_1, \ldots \overset{\text{IID}}{\sim} \mathcal{U}[0,1]^d$$

to approximate $\mu$? Since both sample means are unbiased, we can compare their variances (mean squared errors):

$$\mathrm{var}(\widehat{\mu}_{f,n}) = \frac{\mathrm{var}(f(\boldsymbol{X}))}{n} \qquad \mathrm{var}(\widehat{\mu}_{g,n}) = \frac{\mathrm{var}(g(\boldsymbol{X}))}{n}$$

so it depends on the variances of the two functions

# Keister example

$$\mu = \int_{\mathbb{R}^d} \cos(\|\boldsymbol{t}\|)\exp(-\|\boldsymbol{t}\|^2)\,\mathrm{d}\boldsymbol{t}$$

$$\boldsymbol{x} = \big(\Phi(at_1), \ldots, \Phi(at_d)\big),$$ $\Phi$ is the CDF of the standard normal

$$\mathrm{d}\boldsymbol{x} = a^d \phi(at_1)\cdots\phi(at_d)\,\mathrm{d}\boldsymbol{t} = \frac{a^d \exp(-a^2\|\boldsymbol{t}\|^2/2)}{(2\pi)^{d/2}}\,\mathrm{d}\boldsymbol{t}$$

$$\boldsymbol{t} = \big(\Phi^{-1}(x_1), \ldots, \Phi^{-1}(x_d)\big)/a$$

$$\mu = \int_{[0,1]^d} \frac{(2\pi)^{d/2}}{a^d} \cos\left(\left\|\big(\Phi^{-1}(x_1), \ldots, \Phi^{-1}(x_d)\big)\right\|/a\right)$$

$$\times \exp\left(-\frac{2-a^2}{2a^2}\left\|\big(\Phi^{-1}(x_1), \ldots, \Phi^{-1}(x_d)\big)\right\|^2\right)\mathrm{d}\boldsymbol{x}, \qquad (a^2 \le 2)$$

Whole family $f(\boldsymbol{x}; a)$ for which $\mu = \mathbb{E}[f(\boldsymbol{X}; a)]$ for $\boldsymbol{X} \sim \mathcal{U}[0,1]^d$

# Variable transformations

Suppose that you want to approximate

$$\mu = \int_{\mathscr{T}} g(\boldsymbol{t})\,\mathrm{d}\boldsymbol{t}$$

by Monte Carlo. You need to write it as an expectation. Let $\boldsymbol{t} = \boldsymbol{\psi}(\boldsymbol{x})$, for some invertible transformation, $\boldsymbol{\psi} : \mathscr{X} \to \mathscr{T}$. Define the Jacobian determinant as

$$J(\boldsymbol{x};\boldsymbol{\psi}) := \det\left(\left(\frac{\partial \psi_i}{\partial x_j}\right)_{i,j=1}^{d}\right), \quad J(\boldsymbol{t};\boldsymbol{\psi}^{-1}) := \det\left(\left(\frac{\partial \psi_i^{-1}}{\partial t_j}\right)_{i,j=1}^{d}\right), \; J(\boldsymbol{\psi}(\boldsymbol{x});\boldsymbol{\psi}^{-1}) = \frac{1}{J(\boldsymbol{x};\boldsymbol{\psi})}$$

Then

$$\mu = \int_{\mathcal{X}} g(\boldsymbol{\psi}(\boldsymbol{x}))\,\left|J(\boldsymbol{x};\boldsymbol{\psi})\right|\,\mathrm{d}\boldsymbol{x} = \int_{\mathcal{X}} \underbrace{\frac{g(\boldsymbol{\psi}(\boldsymbol{x}))}{\varrho_{\boldsymbol{X}}(\boldsymbol{x})}\,\left|J(\boldsymbol{x};\boldsymbol{\psi})\right|}_{f(\boldsymbol{x})}\,\varrho_{\boldsymbol{X}}(\boldsymbol{x})\,\mathrm{d}\boldsymbol{x} = \mathbb{E}_{\boldsymbol{X}\sim\varrho_{\boldsymbol{X}}}[f(\boldsymbol{X})]$$

# Variable transformations

$$\mu = \int_{\mathscr{T}} g(\boldsymbol{t})\,\mathrm{d}\boldsymbol{t} = \,?$$

Let $\boldsymbol{t} = \boldsymbol{\psi}(\boldsymbol{x})$, for some invertible transformation, $\boldsymbol{\psi} : \mathscr{X} \to \mathscr{T}$. Then

$$\mu = \int_{\mathcal{X}} g(\boldsymbol{\psi}(\boldsymbol{x}))\,\big|J(\boldsymbol{x};\boldsymbol{\psi})\big|\,\mathrm{d}\boldsymbol{x} = \int_{\mathcal{X}} \underbrace{\frac{g(\boldsymbol{\psi}(\boldsymbol{x}))}{\varrho(\boldsymbol{x})}\,\big|J(\boldsymbol{x};\boldsymbol{\psi})\big|}_{f(\boldsymbol{x})}\,\varrho(\boldsymbol{x})\,\mathrm{d}\boldsymbol{x} = \mathbb{E}_{\boldsymbol{X}\sim\varrho}[f(\boldsymbol{X})]$$

If $\boldsymbol{X} \sim \mathscr{U}[0,1]^d$ (because we know how generate uniform samples), then

$$\mu = \int_{[0,1]^d} \underbrace{g(\boldsymbol{\psi}(\boldsymbol{x}))\,\big|J(\boldsymbol{x};\boldsymbol{\psi})\big|}_{f(\boldsymbol{x})}\,\mathrm{d}\boldsymbol{x} = \mathbb{E}_{\boldsymbol{X}\sim\mathscr{U}[0,1]^d}[f(\boldsymbol{X})]$$

# Transformations of random variables

Let

- $\boldsymbol{T}$ be a random variable with sample space $\mathcal{T}$ and density $\varrho_{\boldsymbol{T}}$
- $\boldsymbol{X}$ be a random variable with sample space $\mathcal{X}$ and density $\varrho_{\boldsymbol{X}}$
- $\boldsymbol{T} = \boldsymbol{\psi}(\boldsymbol{X})$, for some invertible transformation, $\boldsymbol{\psi} : \mathcal{X} \to \mathcal{T}$. Then

$$\varrho_{\boldsymbol{T}}(\boldsymbol{t}) = \varrho_{\boldsymbol{X}}(\boldsymbol{\psi}^{-1}(\boldsymbol{t}))|J(\boldsymbol{t};\boldsymbol{\psi}^{-1})|, \qquad \varrho_{\boldsymbol{T}}(\boldsymbol{\psi}(\boldsymbol{x})) = \varrho_{\boldsymbol{X}}(\boldsymbol{x})|J(\boldsymbol{\psi}(\boldsymbol{x}));\boldsymbol{\psi}^{-1})| = \frac{\varrho_{\boldsymbol{X}}(\boldsymbol{x})}{|J(\boldsymbol{x};\boldsymbol{\psi})|}$$

$$\mu = \int_{\mathcal{T}} \underbrace{h(\boldsymbol{t})\,\varrho_{\boldsymbol{T}}(\boldsymbol{t})}_{g(\boldsymbol{t})}\,\mathrm{d}\boldsymbol{t} = \mathbb{E}_{\boldsymbol{T}\sim\varrho_{\boldsymbol{T}}}[h(\boldsymbol{T})] = \mathbb{E}_{\boldsymbol{X}\sim\varrho_{\boldsymbol{X}}}[h(\boldsymbol{\psi}(\boldsymbol{X}))] \qquad = \int_{\mathcal{X}} h(\boldsymbol{\psi}(\boldsymbol{x}))\,\varrho_{\boldsymbol{X}}(\boldsymbol{x})\,\mathrm{d}\boldsymbol{x}$$

$$\text{or } = \int_{\mathcal{X}} \frac{h(\boldsymbol{\psi}(\boldsymbol{x}))\,\varrho_{\boldsymbol{T}}(\boldsymbol{\psi}(\boldsymbol{x}))}{\varrho_{\boldsymbol{X}}(\boldsymbol{x})}\,|J(\boldsymbol{x};\boldsymbol{\psi})|\,\varrho_{\boldsymbol{X}}(\boldsymbol{x})\,\mathrm{d}\boldsymbol{x} \qquad = \int_{\mathcal{X}} h(\boldsymbol{\psi}(\boldsymbol{x}))\,\varrho_{\boldsymbol{X}}(\boldsymbol{x})\,\mathrm{d}\boldsymbol{x}$$

# Importance sampling

Suppose that you want to approximate

$$\mu = \mathbb{E}_{\boldsymbol{T}\sim\varrho_{\boldsymbol{T}}}[h(\boldsymbol{T})] = \int_{\mathcal{T}} h(\boldsymbol{t})\,\varrho_{\boldsymbol{T}}(\boldsymbol{t})\,\mathrm{d}\boldsymbol{t} = \int_{\mathcal{T}} \underbrace{h(\boldsymbol{t})\,\overbrace{\frac{\varrho_{\boldsymbol{T}}(\boldsymbol{t})}{\varrho_{\boldsymbol{X}}(\boldsymbol{t})}}^{\text{likelihood ratio}}}_{f(\boldsymbol{t})}\,\varrho_{\boldsymbol{X}}(\boldsymbol{t})\,\mathrm{d}\boldsymbol{t} = \mathbb{E}_{\boldsymbol{X}\sim\varrho_{\boldsymbol{X}}}[f(\boldsymbol{X})]$$

We try to choose $\varrho_{\boldsymbol{X}}$ to make $\mathrm{var}[f(\boldsymbol{X})]$ small when doing IID sampling. If $\boldsymbol{T} = \boldsymbol{\psi}(\boldsymbol{X})$ for $\boldsymbol{X} \sim \mathcal{U}[0,1]^d$—since often find generate non-uniform samples from uniform ones—then

$$\mu = \int_{\mathcal{T}} h(\boldsymbol{t})\,\varrho_{\boldsymbol{T}}(\boldsymbol{t})\,\mathrm{d}\boldsymbol{t} = \mathbb{E}_{\boldsymbol{T}\sim\varrho_{\boldsymbol{T}}}[h(\boldsymbol{T})] = \mathbb{E}_{\boldsymbol{X}\sim\mathcal{U}[0,1]^d}[h(\boldsymbol{\psi}(\boldsymbol{X}))] = \int_{[0,1]^d} h(\boldsymbol{\psi}(\boldsymbol{x}))\,\mathrm{d}\boldsymbol{x}$$

# Importance sampling cont'd

Suppose that you want to approximate

$$\mu = \mathbb{E}_{\boldsymbol{T} \sim \varrho_{\boldsymbol{T}}}[h(\boldsymbol{T})] = \int_{\mathcal{T}} h(\boldsymbol{t})\, \varrho_{\boldsymbol{T}}(\boldsymbol{t})\, \mathrm{d}\boldsymbol{t} = \int_{\mathcal{T}} \underbrace{h(\boldsymbol{t})\, \overbrace{\frac{\varrho_{\boldsymbol{T}}(\boldsymbol{t})}{\varrho_{\boldsymbol{X}}(\boldsymbol{t})}}^{\text{likelihood ratio}}}_{f(\boldsymbol{t})}\, \varrho_{\boldsymbol{X}}(\boldsymbol{t})\, \mathrm{d}\boldsymbol{t} = \mathbb{E}_{\boldsymbol{X} \sim \varrho_{\boldsymbol{X}}}[f(\boldsymbol{X})]$$

$$\hat{\mu}_{\mathrm{IS},n} = \frac{1}{n} \sum_{i=0}^{n-1} h(\boldsymbol{X}_i) \frac{\varrho_{\boldsymbol{T}}(\boldsymbol{X}_i)}{\varrho_{\boldsymbol{X}}(\boldsymbol{X}_i)}, \qquad \boldsymbol{X}_0, \boldsymbol{X}_1, \ldots \overset{\mathrm{IID}}{\sim} \varrho_{\boldsymbol{X}}$$

$$\operatorname{var}\left(\hat{\mu}_{\mathrm{IS},n}\right) \approx \frac{1}{n(n-1)} \sum_{i=0}^{n-1} \left[ h(\boldsymbol{X}_i) \frac{\varrho_{\boldsymbol{T}}(\boldsymbol{X}_i)}{\varrho_{\boldsymbol{X}}(\boldsymbol{X}_i)} - \hat{\mu}_{\mathrm{IS},n} \right]^2$$

# Ex. Brownian motion with drift

In computational finance certain outcomes, such as positive option payoffs, may be rare

- Adding a drift to the Brownian motion increases the probability that asset paths yield positive payoffs and
- Reduces the variance of our estimator of the option price (population mean)
- Paths with positive payoffs receive smaller sample weights to compensate and keep the mean the same

$$\text{price} = \int_{\mathbb{R}^d} \text{payoff}(\boldsymbol{x})\, \varrho(\boldsymbol{x})\, \mathrm{d}\boldsymbol{x}, \quad \varrho(\boldsymbol{x}) = \frac{\exp(-\boldsymbol{x}^\top \Sigma^{-1} \boldsymbol{x}/2)}{\sqrt{(2\pi)^d |\Sigma|}}, \;\; \Sigma = \big(\min(t_i, t_j)\big)_{i,j=1}^{d}$$

Here the density $\varrho$ is a discretized Brownian motion

# Ex. Brownian motion with drift cont'd

$$\text{price} = \int_{\mathbb{R}^d} \text{payoff}(\boldsymbol{x})\, \varrho(\boldsymbol{x})\, \mathrm{d}\boldsymbol{x}$$

$$\varrho(\boldsymbol{x}) = \frac{\exp(-\boldsymbol{x}^\top \Sigma^{-1} \boldsymbol{x}/2)}{\sqrt{(2\pi)^d |\Sigma|}}, \quad \Sigma = \big(\min(t_i, t_j)\big)_{i,j=1}^d$$

We importance sample:

$$\varrho_{\text{drift}}(\boldsymbol{x}) = \frac{\exp\big(-(\boldsymbol{x}-\boldsymbol{a})^\top \Sigma^{-1} (\boldsymbol{x}-\boldsymbol{a})/2\big)}{\sqrt{(2\pi)^d |\Sigma|}}, \qquad \boldsymbol{a} = \alpha(t_1, \ldots, t_d)^\top$$

$$\text{likelihood ratio} = \frac{\varrho(\boldsymbol{x})}{\varrho_{\text{drift}}(\boldsymbol{x})} = \exp(-\alpha x_d + \alpha^2 t_d/2)$$

$$\text{price} = \int_{\mathbb{R}^d} \text{payoff}(\boldsymbol{x})\, \exp(-\alpha x_d + \alpha^2 t_d/2)\, \varrho_{\text{drift}}(\boldsymbol{x})\, \mathrm{d}\boldsymbol{x}$$

Note that the weight gets smaller as $\alpha x_d$ increases or $\alpha^2 t_d$ decreases

# Variable transformation / importance sampling observations

- Variable transformation and importance sampling are intertwined
- The choice of sampling density or variable transformation affects the variance of the sample mean
- You must choose the variable transformation so that
  - The Jacobian determinant, $J(\boldsymbol{x};\boldsymbol{\psi})$, is finite
  - The probability density, $\varrho_X(\boldsymbol{x})$, does not vanish unless integrand vanishes
- You might try pilot samples to help choose transformation

# Control variates

$$\mu = \mathbb{E}[f(\boldsymbol{X})] \iff \mu = \mathbb{E}[f(\boldsymbol{X}) - \beta_1\eta_1(\boldsymbol{X}) - \cdots - \beta_m\eta_m(\boldsymbol{X})]$$

provided that $\mathbb{E}[\eta_1(\boldsymbol{X})] = \cdots \mathbb{E}[\eta_m(\boldsymbol{X})] = 0$. Want to arrange such that

$$\mathrm{var}[f(\boldsymbol{X}) - \beta_1\eta_1(\boldsymbol{X}) - \cdots - \beta_m\eta_m(\boldsymbol{X})] \leq \mathrm{var}[f(\boldsymbol{X})]$$

Then

$\hat{\mu}_{\mathrm{CV},n} = \frac{1}{n}\sum_{i=0}^{n-1}[f(\boldsymbol{X}_i) - \beta_1\eta_1(\boldsymbol{X}_i) - \cdots - \beta_m\eta_m(\boldsymbol{X}_i)]$ converges faster than $\hat{\mu}_n = \frac{1}{n}\sum_{i=0}^{n-1} f(\boldsymbol{X}_i)$

$$\mathrm{var}\left(\hat{\mu}_{\mathrm{CV},n}\right) \approx \frac{1}{n(n-1)}\sum_{i=0}^{n-1}\left[f(\boldsymbol{X}_i) - \beta_1\eta_1(\boldsymbol{X}_i) - \cdots - \beta_m\eta_m(\boldsymbol{X}_i) - \hat{\mu}_{\mathrm{CV},n}\right]^2$$

How do we minimize $\mathrm{var}[f(\boldsymbol{X}) - \beta_1\eta_1(\boldsymbol{X}) - \cdots - \beta_m\eta_m(\boldsymbol{X})]$?

# Control variates (cont'd)

$$\hat{\mu}_{\mathrm{CV},n} = \frac{1}{n}\sum_{i=0}^{n-1}[f(\boldsymbol{X}_i) - \beta_1\eta_1(\boldsymbol{X}_i) - \cdots - \beta_m\eta_m(\boldsymbol{X}_i)]$$

How do we minimize $\mathrm{var}[f(\boldsymbol{X}) - \beta_1\eta_1(\boldsymbol{X}) - \cdots - \beta_m\eta_m(\boldsymbol{X})]$? Use least squares regression with centered response and explanatory variables to choose $\hat{\boldsymbol{\beta}}$:

$$\hat{\boldsymbol{\beta}} = \operatorname*{argmin}_{\boldsymbol{b}} \|\boldsymbol{y} - \mathsf{H}\boldsymbol{b}\|^2$$

where $\boldsymbol{y} = \begin{pmatrix} f(\boldsymbol{x}_0) - \widehat{\mu}_n \\ \vdots \\ f(\boldsymbol{x}_{n-1}) - \widehat{\mu}_n \end{pmatrix}, \quad \mathsf{H} = \begin{pmatrix} \eta_1(\boldsymbol{x}_0) - \widehat{\mu}_{\eta_1,n} & \cdots & \eta_m(\boldsymbol{x}_0) - \widehat{\mu}_{\eta_m,n} \\ \vdots & & \vdots \\ \eta_1(\boldsymbol{x}_{n-1}) - \widehat{\mu}_{\eta_1,n} & \cdots & \eta_m(\boldsymbol{x}_{n-1}) - \widehat{\mu}_{\eta_m,n} \end{pmatrix}$

$$\widehat{\mu}_n = \frac{1}{n}\sum_{i=0}^{n-1} f(\boldsymbol{x}_i), \qquad \widehat{\mu}_{\eta_1,n} = \frac{1}{n}\sum_{i=0}^{n-1}\eta_1(\boldsymbol{x}_i), \; \ldots, \; \widehat{\mu}_{\eta_m,n} = \frac{1}{n}\sum_{i=0}^{n-1}\eta_m(\boldsymbol{x}_i)$$

# Least Squares

How do we solve

$$\hat{\boldsymbol{\beta}} = \underset{\boldsymbol{b}}{\operatorname{argmin}} \left\| \boldsymbol{y} - \mathsf{H}\boldsymbol{b} \right\|^2 = \underset{\boldsymbol{b}}{\operatorname{argmin}} (\boldsymbol{y} - \mathsf{H}\boldsymbol{b})^T (\boldsymbol{y} - \mathsf{H}\boldsymbol{b})$$

Let

$$\boldsymbol{b} = (\mathsf{H}^\top \mathsf{H})^{-1} \mathsf{H}^\top \boldsymbol{y} + \boldsymbol{c}$$

Then

$$\begin{aligned}
\left\| \boldsymbol{y} - \mathsf{H}\boldsymbol{b} \right\|^2 &= \left\| \boldsymbol{y} - \mathsf{H}[(\mathsf{H}^\top \mathsf{H})^{-1} \mathsf{H}^\top \boldsymbol{y} + \boldsymbol{c}] \right\|^2 \\
&= \left\| [\mathsf{I} - \mathsf{H}(\mathsf{H}^\top \mathsf{H})^{-1} \mathsf{H}^\top] \boldsymbol{y} + \mathsf{H}\boldsymbol{c} \right\|^2 \\
&= \left( [\mathsf{I} - \mathsf{H}(\mathsf{H}^T \mathsf{H})^{-1} \mathsf{H}^\top] \boldsymbol{y} + \mathsf{H}\boldsymbol{c} \right)^\top \left( [\mathsf{I} - \mathsf{H}(\mathsf{H}^\top \mathsf{H})^{-1} \mathsf{H}^\top] \boldsymbol{y} + \mathsf{H}\boldsymbol{c} \right) \\
&= \boldsymbol{y}^\top [\mathsf{I} - \mathsf{H}(\mathsf{H}^\top \mathsf{H})^{-1} \mathsf{H}^\top] \boldsymbol{y} + \boldsymbol{c}^\top \mathsf{H}^\top \mathsf{H}\boldsymbol{c}
\end{aligned}$$

So the optimum comes by setting $\boldsymbol{c} = \boldsymbol{0}$

# Control variates

- Should be clever in choosing the $\eta_j$
  - Good choices can decrease the sample size required
  - Bad choices only hurt in terms of increased computation time, but not increased sample size
- The $\beta_j$ should be chosen by least squares for IID sampling
- For low discrepancy sequences, the $\beta_j$ should be chosen to reduce the variation (not variance)

# Conditional Monte Carlo

Recall that

$$\mathbb{E}(Y) = \mathbb{E}_{\boldsymbol{Z}}[\mathbb{E}_Y(Y|\boldsymbol{Z})], \quad \text{var}(Y) = \mathbb{E}_{\boldsymbol{Z}}[\text{var}(Y|\boldsymbol{Z})] + \text{var}_{\boldsymbol{Z}}[\mathbb{E}_Y(Y|\boldsymbol{Z})]$$

Which implies that

$$\text{var}_{\boldsymbol{Z}}[\mathbb{E}_Y(Y|\boldsymbol{Z})] \leq \text{var}(Y), \qquad \text{var}\left(\frac{1}{n}\sum_{i=0}^{n-1}\mathbb{E}_Y(Y|\boldsymbol{Z}_i)\right) \leq \text{var}\left(\frac{1}{n}\sum_{i=0}^{n-1}Y_i\right)$$

Let $(Y = f(\boldsymbol{X}_{1:d}), \boldsymbol{X}_{1:d})$ be be a vector of random variables

Let $Z = \boldsymbol{X}_{2:d}$

If $g(\boldsymbol{X}_{2:d}) = \mathbb{E}_Y(Y \mid \boldsymbol{X}_{2:d})$ has an analytic form, then

$\mu = \mathbb{E}(Y) = \mathbb{E}_{\boldsymbol{X}_{2:d}}[g(\boldsymbol{X}_{2:d})]$ and $\text{var}(Y) \geq \text{var}_{\boldsymbol{X}_{2:d}}[g(\boldsymbol{X}_{2:d})]$

If $h(y, \boldsymbol{X}_{2:d}) = \varrho_{Y|\boldsymbol{X}_{2:d}}(y \mid \boldsymbol{x}_{2:d})$ has an analytic form, then

$\varrho_Y(y) = \mathbb{E}_{\boldsymbol{X}_{2:d}}[h(y, \boldsymbol{X}_{2:d})]$ probability density expressed as expectation

# Conditional Monte Carlo for Density Estimation

Estimate the probability density, $\varrho$, for $Y = f(X)$, where $\boldsymbol{X} \sim \mathcal{U}[0,1]^d$

If one can identify an increasing $g$, such that

$$y = f(\boldsymbol{x}) \iff x_1 = g(y; \boldsymbol{x}_{2:d})$$

Then

$$\begin{aligned} F_{Y|\boldsymbol{X}_{2:d}}(y \,|\, \boldsymbol{x}_{2:d}) &= \mathbb{P}[Y \le y \,|\, \boldsymbol{X}_{2:d} = \boldsymbol{x}_{2:d}] = \mathbb{P}[f(\boldsymbol{X}) \le y \,|\, \boldsymbol{X}_{2:d} = \boldsymbol{x}_{2:d}] \\ &= \mathbb{P}[X_1 \le g(y; \boldsymbol{x}_{2:d})] = g(y; \boldsymbol{x}_{2:d}) \end{aligned}$$

$$\varrho(y) = \mathbb{E}\left[\varrho_{Y|\boldsymbol{X}_{2:d}}(y \,|\, \boldsymbol{x}_{2:d})\right] = \int_{[0,1]^{d-1}} g'(y; \boldsymbol{x})\, \mathrm{d}\boldsymbol{x}$$

# Ex. Density of a sum of uniforms

Let $Y = \gamma_1 X_1 + \cdots + \gamma_d X_d$, where $X \sim \mathcal{U}[0,1]^d$ and all the $\gamma_\ell$ are non-negative. What is the PDF of $Y$?

$$g(y; x_2, \ldots, x_d) = \frac{1}{\gamma_1}(y - \gamma_2 x_2 - \cdots - \gamma_d x_d), \quad \gamma_2 x_2 + \cdots + \gamma_d x_d \le y \le \gamma_1 + \gamma_2 x_2 + \cdots + \gamma_d x_d$$

$$g'(y; x_2, \ldots, x_d) = \frac{1}{\gamma_1}, \quad \gamma_2 x_2 + \cdots + \gamma_d x_d \le y \le \gamma_1 + \gamma_2 x_2 + \cdots + \gamma_d x_d$$

# Antithetic Sampling

If you are sampling from a probability distribution with symmetry, then you can evaluate $f$ at both $X_i$ and $T(X_i)$, where $T(X)$ has the same distribution as $X$, e.g.,

$$T(X) = \mathbf{1} - X \text{ for } X \sim \mathcal{U}[0,1]^d \text{ or } T(X) = -X \text{ for } X \sim \mathcal{N}(\mathbf{0}, \Sigma)$$

Note that if $X_i$ are IID, then $X_i$ and $T(X_i)$ are not independent, but $\big(X_i, T(X_i)\big)$ are

When does $\hat{\mu}_{\text{Anti},n} = \dfrac{1}{2n} \displaystyle\sum_{i=0}^{n-1} [f(X_i) + f(T(X_i))]$ have a smaller variance than $\hat{\mu}_{2n} = \dfrac{1}{2n} \displaystyle\sum_{i=0}^{2n-1} f(X_i)$ for IID $X_i$?

# Antithetic Sampling

Note that

$$\mathrm{var}(\hat{\mu}_{\mathrm{Anti},n}) = \frac{1}{4n^2}\sum_{i=0}^{n-1}\mathrm{var}\big(f(\boldsymbol{X}_i)+f(\boldsymbol{T}(\boldsymbol{X}_i))\big) = \frac{\mathrm{var}\big(f(\boldsymbol{X})\big)}{2n}\left[1+\mathrm{corr}\big(f(\boldsymbol{X}),f(\boldsymbol{T}(\boldsymbol{X}))\big)\right]$$

Whereas

$$\mathrm{var}(\hat{\mu}_{2n}) = \frac{\mathrm{var}\big(f(\boldsymbol{X})\big)}{2n}$$

So antithetic sampling helps if $\mathrm{corr}\big(f(\boldsymbol{X}),f(\boldsymbol{T}(\boldsymbol{X}))\big) < 0$

$$\mathrm{var}(\hat{\mu}_{\mathrm{Anti},n}) \approx \frac{1}{4n(n-1)}\sum_{i=0}^{n-1}\left[f(\boldsymbol{X}_i)+f(\boldsymbol{T}(\boldsymbol{X}_i)) - 2\hat{\mu}_{\mathrm{Anti},n}\right]^2$$

# How to get an antithetic variate

Suppose that you have random variable $X$ with CDF $F$. One way to construct an antithetic variate, $T(X)$:

- Note that $U = F(X) \sim \mathcal{U}[0,1]$
- Also, $1 - U \sim \mathcal{U}[0,1]$
- Therefore $T(X) = F^{-1}(1 - U) = F^{-1}\big(1 - F(X)\big)$ also has CDF $F$

# Latin Hypercube Sampling in One Dimension

Let's be intentional about placing uniform looking points on $[0,1]$

$$X_i = (i + U_i)/n, \quad U_0, U_1, \ldots \overset{\text{IID}}{\sim} \mathcal{U}[0,1],\ i = 0,\ldots,n-1, \qquad \hat{\mu}_{\text{LHS},n} = \frac{1}{n}\sum_{i=0}^{n-1} f(X_i)$$

Note that $X_i = (i + U_i)/n \sim \mathcal{U}[i/n, (i+1)/n]$, and so

- $$\mathbb{E}\left[\hat{\mu}_{\text{LHS},n}\right] = \frac{1}{n}\sum_{i=0}^{n-1}\mathbb{E}[f(X_i)] = \frac{1}{n}\sum_{i=0}^{n-1}\int_{i/n}^{(i+1)/n} f(x)\times n \times \mathrm{d}x = \int_0^1 f(x)\mathrm{d}x = \mu$$

  which means that $\hat{\mu}_{\text{LHS},n}$ is unbiased

- The $X_i$ are not IID but they are independent

# Variance of LHS in One Dimension

$$\operatorname{var}(\widehat{\mu}_{\mathrm{LHS},n}) = \operatorname{var}\left(\sum_{i=0}^{n-1} \frac{f(X_i)}{n}\right) = \frac{1}{n^2}\sum_{i=0}^{n-1} \operatorname{var}\bigl(f(X_i)\bigr)$$ since the $X_i$ are uncorrelated

$$= \frac{1}{n^2}\sum_{i=0}^{n-1} \int_{i/n}^{(i+1)/n} \bigl(f(x) - f(\eta_i)\bigr)^2 \times n \times \mathrm{d}x$$

where $f(\eta_i) = \mathbb{E}[f(X_i)]$ for some $\eta_i$ by the Mean Value Theorem

$$= \frac{1}{n}\sum_{i=0}^{n-1} \int_{i/n}^{(i+1)/n} \bigl(f'(\xi_i(x))(x-\eta_i)\bigr)^2 \mathrm{d}x,$$

where $f(x) = f(\eta_i) + f'(\xi_i(x))(x-\eta_i)$ for some $\xi(x)$ by Taylor's Theorem

$$\le \frac{1}{n}\sum_{i=0}^{n-1} \frac{\|f'\|_\infty^2}{n^3} = \frac{\|f'\|_\infty^2}{n^3}$$

$$\operatorname{rmse}(\widehat{\mu}_{\mathrm{LHS},n}) \le \frac{\|f'\|_\infty}{n^{3/2}}$$

# Latin Hypercube Sampling in $d$ dimensions

$X_{i,\ell} = (\pi_\ell(i) + U_{i,l})/n, \quad U_{i,\ell} \overset{\text{IID}}{\sim} \mathcal{U}[0,1]$, $\pi_\ell$ are IID random permutations of $1,\ldots,n$

$$\hat{\mu}_{\text{LHS},n} = \frac{1}{n}\sum_{i=0}^{n-1} f(X_{i,1}, \ldots X_{i,d})$$

To quantify uncertainty in the estimate of the mean we need replications:

$X^{(r)}_{i,\ell} = (\pi^{(r)}_\ell(i) + U^{(r)}_{i,\ell})/n$, $U^{(r)}_{i,\ell} \overset{\text{IID}}{\sim} \mathcal{U}[0,1]$, $\pi^{(r)}_\ell$ IID, $\hat{\mu}^{(r)}_{\text{LHS},n} = \frac{1}{n}\sum_{i=0}^{n-1} f(X^{(r)}_{i,1}, \ldots X^{(r)}_{i,d})$, $r = 1,\ldots,R$

$$\hat{\mu}_{\text{LHS},n,R} = \frac{1}{R}\sum_{r=0}^{R-1} \hat{\mu}^{(r)}_{\text{LHS},n} = \frac{1}{nR}\sum_{r,i=0}^{R-1,n-1} f(X^{(r)}_{i,1}, \ldots X^{(r)}_{i,d})$$

$$\text{var}\left(\hat{\mu}_{\text{LHS},n,R}\right) \approx \frac{1}{R(R-1)}\sum_{r=0}^{R-1}\left[\hat{\mu}^{(r)}_{\text{LHS},n} - \hat{\mu}_{\text{LHS},n,R}\right]^2$$

# Pros and cons of LHS

## Pros

- Do not need to guess importance sampling distributions or control variates or compute conditional expectations
- Can give faster convergence rates than IID

## Cons

- Sample size, $n$, is fixed
- Difficult to assess accuracy, except by replications; too many replications destroy the faster convergence
- Great convergence rates do not hold for multivariate case ($d > 1$)

# Low discrepancy sampling

Low discrepancy sampling, also known as quasi-Monte Carlo methods is an improvement on LHS that can provide higher order convergence rates than IID sampling for functions with modest smoothness

Recall from earlier the discrepancy, $D\left(\{\boldsymbol{x}_i\}_{i=0}^{n-1}, F; K\right)$, between a point cloud or node set $\{\boldsymbol{x}_i\}_{i=0}^{n-1}$ and a probability distribution, $F$, defined by

$$D^2\left(\{\boldsymbol{x}_i\}_{i=0}^{n-1}, F; K\right) = \int_{\mathscr{X}\times\mathscr{X}} K(\boldsymbol{x},\boldsymbol{x}')\,\mathrm{d}F(\boldsymbol{x})\mathrm{d}F(\boldsymbol{x}') - \frac{2}{n}\sum_{i=0}^{n-1}\int_{\mathscr{X}} K(\boldsymbol{x},\boldsymbol{x}_i)\,\mathrm{d}F(\boldsymbol{x})$$

$$+\frac{1}{n^2}\sum_{i,j=0}^{n-1} K(\boldsymbol{x}_i,\boldsymbol{x}_j),$$

Where we will take the domain to be $\mathscr{X} = [0,1]^d$

# Discrepancy interpretation

- Worst-case error

$$\left| \int_{[0,1]^d \times [0,1]^d} f(\boldsymbol{x})\,\mathrm{d}(\boldsymbol{x}) - \frac{1}{n}\sum_{i=0}^{n-1} f(\boldsymbol{z}_i) \right| \le D\left(\mathscr{U}[0,1]^d, \{\boldsymbol{z}_i\}_{i=0}^{n-1}; K\right) \|f\|_{\mathscr{H}} \quad \forall f \in \mathscr{H}$$

- Average-case error

$$\mathbb{E}_{f \in \mathscr{GP}(0,K)} \left| \int_{[0,1]^d} f(\boldsymbol{x})\,\mathrm{d}\boldsymbol{x} - \frac{1}{n}\sum_{i=0}^{n-1} f(\boldsymbol{x}_i) \right|^2 = D^2\left(\mathscr{U}[0,1]^d, \{\boldsymbol{x}_i\}_{i=0}^{n-1}\right)$$

- Distance between the target distribution and the point cloud

$$\mathbb{E}_{\boldsymbol{X},\boldsymbol{X}' \sim \mathcal{U}[0,1]^d} K(\boldsymbol{X},\boldsymbol{X}') - 2\mathbb{E}_{\boldsymbol{X} \sim \mathcal{U}[0,1]^d, \boldsymbol{Z} \sim F_{\{\boldsymbol{x}_i\}_{i=0}^{n-1}}} K(\boldsymbol{X},\boldsymbol{Z}) + \mathbb{E}_{\boldsymbol{Z},\boldsymbol{Z}' \sim F_{\{\boldsymbol{x}_i\}_{i=0}^{n-1}}} K(\boldsymbol{Z},\boldsymbol{Z}')$$
$$= D^2\left(\mathcal{U}[0,1]^d, \{\boldsymbol{x}_i\}_{i=0}^{n-1}; K\right)$$

# van der Corput sequence

In one-dimension, evenly spaced points may be constructed as

$$x_i = i/n, \quad i = 0,\ldots,n-1$$

But you need to fix $n$ in advance. The van der Corput sequence is an infinite sequence that fills the unit interval $[0,1]$ evenly and one can stop at any $n$ and even keep going. Let $b$ (normally $b = 2$) be a prime base. Then

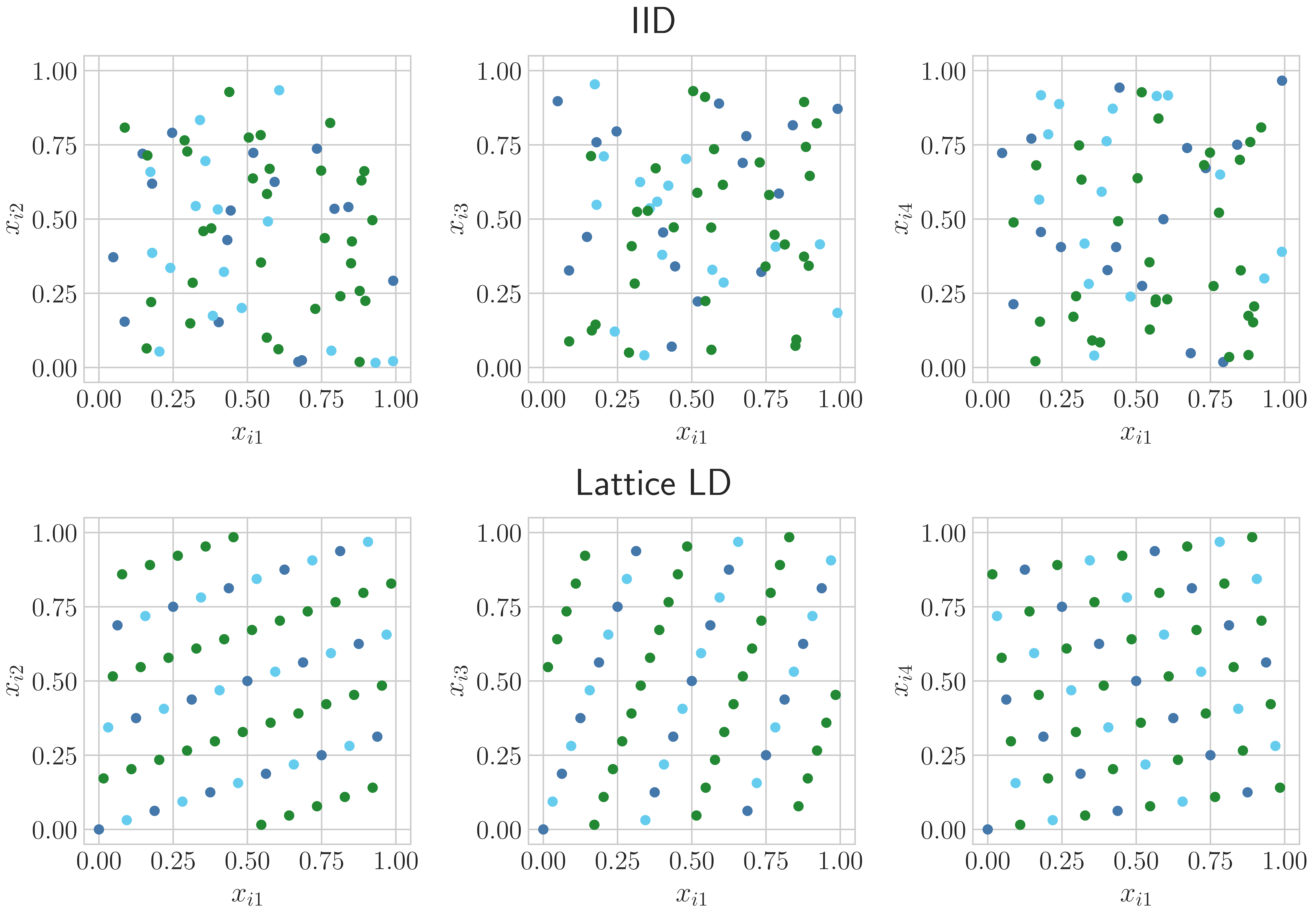
$$\phi_b(i) = \phi_b\left(\cdots + i_2 b^2 + i_1 b + i_0\right) = \phi_b\left((\cdots i_2 i_1 i_0)_b\right) := {}_b(0.i_0 i_1 i_2 \cdots) = \frac{i_0}{b} + \frac{i_1}{b^2} + \frac{i_2}{b^3} + \cdots$$

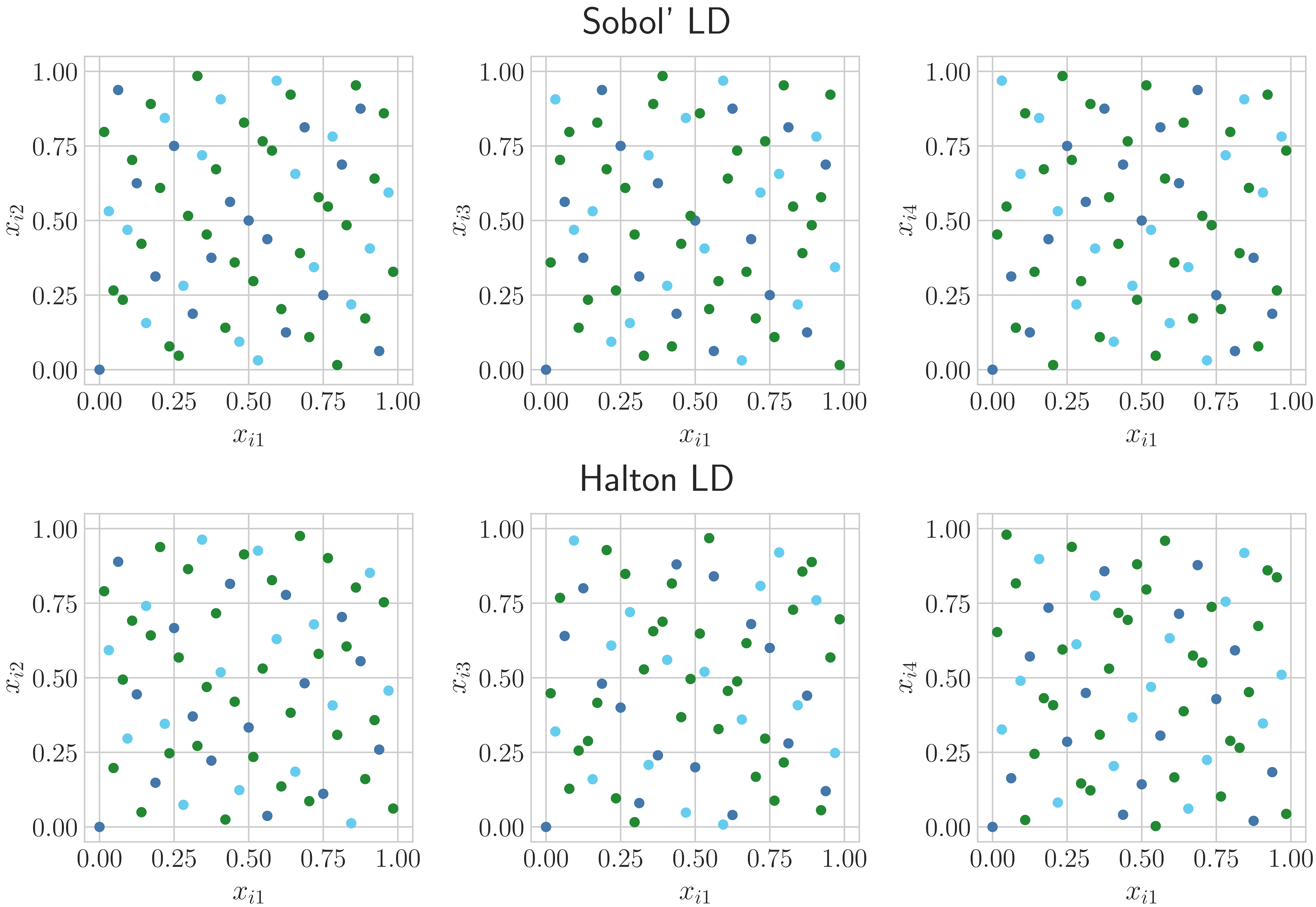
The set $\{i/b^m\}_{i=0}^{b^m-1}$ and the set $\{\phi_b(i)\}_{i=0}^{b^m-1}$ have the same points but in a different order, but the elements of the second set do not change their order when $m$ increases

# Four families of low discrepancy sequences

For $i = 0, 1, \ldots$

- Kronecker: $\boldsymbol{x}_i = i\boldsymbol{\zeta} \bmod \boldsymbol{1}$ for some specially chosen generator $\boldsymbol{\zeta} \in [0,1)^d$, no preferred sample size, $n$
- Lattice: $\boldsymbol{x}_i = \phi_b(i)\boldsymbol{\zeta} \bmod \boldsymbol{1}$ for some specially chosen generator $\boldsymbol{\zeta} \in \mathbb{N}^d$, preferred sample size is $n = b^m$
- Halton: $\boldsymbol{x}_i = \big(\phi_2(i), \phi_3(i), \phi_5(i), \ldots\big)$, essentially no preferred sample size, $n$
- Digital: $\boldsymbol{x}_i = i_0\boldsymbol{c}_0 \oplus i_1\boldsymbol{c}_1 \oplus i_2\boldsymbol{c}_2 \oplus \cdots$, where $i = (\cdots i_2 i_1 i_0)_b$, $\oplus$ means digitwise addition modulo $b$, and $\boldsymbol{c}_0, \boldsymbol{c}_1, \boldsymbol{c}_2, \ldots \in [0,1)^d$, preferred sample size is $n = b^m$


IID
Lattice LD
$x_{i1}$
$x_{i2}$
$x_{i3}$
$x_{i4}$
0.00
0.25
0.50
0.75
1.00



Sobol' LD
Halton LD
1.00
0.75
0.50
0.25
0.00
$x_{i1}$
$x_{i2}$
$x_{i3}$
$x_{i4}$

# Component-by-component search

The generators for digital nets may be chosen by number theory, but in most cases the generators are chosen by computer search to minimize the discrepancy.

The component-by-component search method

1. Choose a generator for a 1-D sequence that minimizes the discrepancy over a range of $n$, set $\ell = 1$
2. Freeze the first $\ell$ dimensions of the generator
3. Choose the $\ell + 1$ component of the generator by minimizing the discrepancy
4. Increases $\ell$ by one
5. If $\ell < d$, go to step 2; otherwise you are done

# Randomizing low discrepancy sequences

Want to make $\boldsymbol{x}_{\text{rand},i} \sim \mathcal{U}[0,1]^d$, while $\{\boldsymbol{x}_{\text{rand},i}\}_{i=0}^{n-1}$ is low discrepancy

- Random shifts: $\boldsymbol{x}_{\text{rand},i} = \boldsymbol{x}_i + \boldsymbol{\Delta} \bmod \mathbf{1}, \quad \boldsymbol{\Delta} \sim \mathcal{U}[0,1]^d$, especially for lattices and Kronecker
- Random digital shifts: $\boldsymbol{x}_{\text{rand},i} = \boldsymbol{x}_i \oplus \boldsymbol{\Delta} \bmod \mathbf{1}, \quad \boldsymbol{\Delta} \sim \mathcal{U}[0,1]^d$, especially for digital sequences
- Linear matrix scrambling or nested uniform scrambling, especially for digital sequences

# Benefits of randomizing

- Unbiased estimator, $\widehat{\mu} = \frac{1}{n}\sum_{i=0}^{n-1} f(\boldsymbol{x}_{\text{rand},i})$, of $\mathbb{E}\left[f(\boldsymbol{X})\right]$ for $\boldsymbol{X} \sim \mathscr{U}[0,1]^d$
- $\boldsymbol{x}_{\text{rand},i}$ almost surely never on the boundary of $\mathscr{U}[0,1]^d$, so transformations to the Gaussian distribution will be finite
- Root mean squared error for digital sequences with linear matrix or nested uniform scrambling can be nearly $\mathscr{O}(n^{-3/2})$